[B19 EC 3101]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech I Semester (R19) Regular Examinations

Linear ICs and Pulse Circuits

DEPARTMENT OF ECE

MODEL QUESTION PAPER

TIME: 3 Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

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			CO	KL	M
		UNIT – I			
1.	a).	Justify that a high pass RC circuit with a small-time constant acts as a differentiator?	1	3	7
	b).	Explain the operation of Double Clippers with and without bias.	1	3	8
		OR			
2.	a).	Explain the response of a low Pass RC circuit to Square input using relevant diagrams and waveforms. Obtain the expression for its output voltage.	1	3	8
	b).	State and prove clamping circuit theorem.	1	3	7
		UNIT – II			
3.	a).	Design astable multivibrator with following specifications. Vcc = Vbb=10V, Ic(sat) =5mA,hfe(Min)= 25, and Maximum Trigger frequency of 25KHz.	2	4	8
	b).	Explain the operation of a Collector Coupled fixed bias Bi-stable Multivibrator Using relevant Diagrams.	2	3	7
		OR			
4.	a).	Explain Switching timings of a transistor.	2	3	7
	b).	Analyze the operation of a Collector Coupled Monostable Multivibrator Using relevant circuit diagrams and waveforms.	2	4	8
		UNIT – III			
5.	a).	Explain the operation of the square wave generator with a neat circuit diagram and derive an expression for its time period.	3	2	8
	b).	With a neat sketch, explain the operation of the Instrumentation amplifier in detail.	3	2	7
		OR			
6.	a).	Design a schmitt circuit with +Vsat=+12V, -Vsat=-12V, R1=50 ohm, R2=100k ohm, Vref=0.2V, Vin=1V(p-p).Find values of V_{utp} and V_{ltp}	3	4	8
	b).	Explain in detail about the operation of Absolute value output circuit with necessary derivations.	3	2	7
		UNIT – IV			

7.	a).	Design a wide BPF with f_L=200Hz ,f_H=1KHz and pass band gain=4. Draw the frequency response plot of this filter.Calculate the value Q for the filter.	4	4	8
	b).	Design an RC phase oscillator for frequency of 5KHz	4	4	7
OR					
8.	a).	Derive the transfer function of a second order LPF. Comment on its frequency response	4	4	7
	b).	Explain the operation of the Quadrature Oscillator with a neat diagram and design the Quadrature Oscillator to operate at a frequency of 159 Hz.	4	4	8
UNIT – V					
9.	a).	Explain the operation of 555 Timer as Monostable Multivibrator with a neat circuit diagram and derive an expression for its time period. Design a Monostable Multivibrator using 555timer to produce a pulse width of 100ms	5	2	9
	b).	Explain in detail about the block diagram of PLL	5	2	6
OR					
10.	a).	Design an Astable Multivibrator using 555timer to generate duty cycle of a) 45% and b) 65%.	5	4	8
	b).	Explain in detail about two applications of PLL	5	2	7

[B19 EC 3102]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech I Semester (R19) Regular Examinations

DIGITAL COMMUNICATION

DEPARTMENT OF ECE

MODEL QUESTION PAPER

TIME: 3 Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

*****

			CO	KL	M
UNIT - I					
1.	a).	Explain about delta modulation (DM) systems and design an Adaptive Delta Modulation system to eliminate the drawbacks generated in the DM system.	**1**	**2**	**8**
	b).	Define quantization. Consider that the signal $cos2\pi t$ is quantized into 16 levels. The sampling rate is 4Hz. Assume that the sampling signal consists of pulses each having a unit height and duration dt. & the pulses occur every $t=k/4sec, \quad -\infty < k < \infty$. a. Sketch the binary signal representing each sample voltage. b. How many bits are required per sample.	**1**	**3**	**7**
OR					
2.	a).	Explain about the operation of a PCM system. What is companding?	**1**	**2**	**8**
	b).	What are the drawbacks in the delta modulation system and how can they be eliminated using CVSD?	**1**	**3**	**7**
UNIT - II					
3.	a).	Explain how a binary signal can be transmitted and received by using a BPSK system?	**2**	**2**	**7**
	b).	With the help of an example, explain the method of generating a phase continuous MSK signal.	**2**	**2**	**8**
OR					
4.	a).	Explain the role of a QPSK transmitter and receiver in serial data transmission and reception	**2**	**2**	**8**

	b).	In a DPSK Receiver, the received bit sequence b(t) is 01101100 then i) Find reconstructed bit sequence d(t) ii) Due to presence of noise b(t) is recovered as 01111100 then detect d(t) and identify the bits which are wrongly detected. Use EX-OR logic.	**2**	**3**	**7**
		UNIT - III			
5.	a).	Explain about linear filtering and calculate noise power output of RC low pass filter, differentiator and an integrator	**3**	**2**	**8**
	b).	Determine the power spectral density of either $n_c(t)$ or $n_s(t)$.	**3**	**3**	**7**
		OR			
6.	a).	Derive expressions for quadrature components of noise.Explain narrow band representation of noise.	**3**	**3**	**8**
	b).	Derive the relation for output noise power spectral density as $G_{no}(f) = \|H(f)\|^2.G_{ni}(f)$ where $G_{ni}(f)$ is the input noise power spectral density of a filter with transfer function H(f).	**3**	**3**	**7**
		UNIT - IV			
7.	a).	What is the function of a baseband signal receiver and derive its probability of error ?	**4**	**3**	**8**
	b).	By deriving expression for Pe for BPSK and BFSK systems, compare the performance of these two data transmission systems.	**4**	**3**	**7**
		OR			
8.	a).	Derive the expressions for Probability of error Pe and transfer function H(f) for an optimum filter	**4**	**3**	**8**
	b).	A signal is either $S1(t) = A \cos 2\pi fot$ or $S2(t) = 0$ for an interval T= n/fo ,with n being an integer. Find the error probability Pe of the matched filter.	**4**	**3**	**7**
		UNIT - V			
9.	a).	Derive the expressions for output SNR when the binary signal is transmitted using BPSK in a PCM system	**5**	**3**	**8**
	b).	Compare the noise performance of PCM and DM systems.	**5**	**2**	**7**
		OR			
10.	a).	How a CDMA system uses spread spectrum to provide multiple access communication.	**5**	**2**	**7**
	b).	Explain the methods of generating PN sequence.	**5**	**2**	**8**

[B19 EC 3103]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech I Semester (R19) Regular Examinations

ANTENNAS AND PROPAGATION

DEPARTMENT OF ECE

MODEL QUESTION PAPER

TIME: 3 Hrs. **Max. Marks: 75M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

*****

			CO	KL	M
UNIT – I					
1.	a).	Derive expressions for the EM fields radiated by an alternating current element.	1	K4	10
	b).	Compute the directivity of a current element.	1	K3	5
OR					
2.	a).	Compute the radiation resistance of a half wave dipole.	1	K3	10
	b).	Explain about different types of current distribution on linear antennas.	1	K2	5
UNIT – II					
3.	a).	What are broadside and end-fire arrays ? Obtain expressions for BWFN and HPBW in both cases.	3	K4	10
	b).	A broadside array has a BWFN of 60deg. Compute the BWFN in degrees if the same array is excited in end-fire fashion.	3	K3	5
OR					
4.	a).	Show that in a uniform linear array, the first side lobe is down the principal maximum by 13.5dB.	3	K3	10
	b).	Explain the technique of pattern multiplication with examples.	3	K2	5
UNIT – III					
5.	a).	With a neat diagram, explain the operating principles of rhombic antenna. List out the disadvantages of a rhombic antenna.	2	K2	10
5	b).	Compute the optimum values of length, height and tilt angle of a rhombic antenna if the angle of elevation alpha at which the EM energy leaves the antenna is 30deg.	2	K3	5
OR					
6.	a).	Explain in detail about the various feed mechanisms of antennas with parabolic reflectors.	2	K2	7
	b).	Derive an expression for the expression of the impedance of the slot antenna.	2	K4	8
UNIT – IV					
7.	a).	Explain in detail about the slotted line method of antenna input impedance measurement.	4	K2	8
	b).	Explain the method of measuring the radiation pattern of an antenna.	4	K2	7

OR					
8.	a).	Explain one method of measurement of antenna gain.	4	K2	8
	b).	Explain the method of measurement of polarization of an antenna.	4	K2	7
UNIT – V					
9.	a).	Derive an expression for the refractive index of the ionosphere.	5	K4	8
	b).	Explain ground wave propagation in detail.	5	K2	7
OR					
10.	a).	Derive an expression for the field strength of a space wave.	5	K4	7
	b).	Explain the terms critical frequency, MUF and skip distance.	5	K2	8

[B19 EC 3104]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech I Semester (R19) Regular Examinations

CONTROL SYSTEMS

DEPARTMENT OF ECE

MODEL QUESTION PAPER

FOR ECE

TIME: 3 Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

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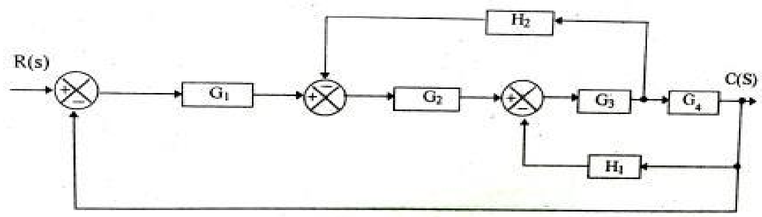
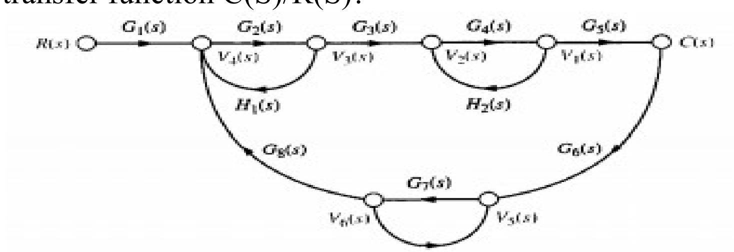
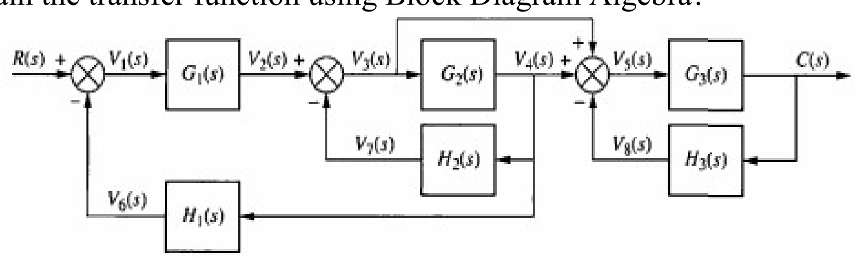
			CO	KL	M
		UNIT – I			
1.	a).	Obtain the transfer function using Block Diagram Algebra? 	1	3	7
	b).	Find the transfer function C(S)/R(S)? 	1	3	8
		OR			
2.	a).	Obtain the transfer function using Block Diagram Algebra? 	1	3	7
	b).	Find the transfer functions of the given differential equation with all initial conditions are zero. $$\frac{d^2y}{dt^2} + 12\frac{dy}{dt} + 32y = 32u(t)$$	1	3	8
		UNIT – II			
3.	a).	Evaluate the steady state error for input $(1+t+t^2)$ u(t) for a unity feedback control system whose open loop transfer function is G(s) =10/(s(s+2)).	2	4	7
	b).	A unity feedback control system has G(s) = 10/s(s+2) If it is subjected to step input of 12 units. Determine:	2	3	8

		(i) Rise time (ii) Maximum peak overshoot			
		OR			
4.	a).	Obtain the response of unity feedback system with open loop transfer function is G(S) = 4/(S(S+5)) when input is step.	2	4	7
	b).	The unity feedback system is characterized by an open loop transfer function G(S) = K/S(S+1), determine the gain K, so that the system will have a damping ratio of 0.5, and also determine (i) Maximum peak overshoot(ii) Peak time	2	3	8
		UNIT – III			
5.	a).	A unity feedback control system has an open loop transfer function G(s)=K/S(S+2)(S+4) Sketch the root locus?	3	2	7
	b).	The characteristic equation of a system is $9S^5-20S^4+10S^3-S^2-9S-10=0$, construct the routh array and comment on the location of the roots.	3	3	8
		OR			
6.	a).	A unity feedback control system has an open loop transfer function $G(s)=K(S^2+6S+25)/(S(S+1)(S+2))$ Sketch the root locus?	3	3	7
	b).	The characteristic equation of a system is $s^5+s^4+2s^3+2s^2+3s+5=0$, construct the routh array and comment on the location of the roots.	3	3	8
		UNIT – IV			
7.	a).	Explain in detail about Nyquist plot with principle of arguments.	4	2	7
	b).	Draw the bode plot for the following forward path gain of a unity feedback control system, $G(S) = 10(S+5)^2/(S^2(S+2)(S+10))$.	4	2	8
		OR			
8.	a).	Sketch the polar plot for the system represented by $G(s)H(s)=1/(S(S+1)^2)$?	4	3	7
	b).	Draw the bode plot for the following forward path gain of a unity feedback control system, G(S) = 10/(S(0.4S+1)(0.1S+1)).	4	3	8
		UNIT – V			
9.	a).	Derive the transfer function equation from the state model equations X=Ax+Bu, Y=Cx+Du.	5	3	7
	b).	Obtain the state model of the system whose transfer function is given as; $y(S)/U(S) = (10S+5)/(S^3+6S^2+7S+8)$	5	3	8
		OR			
10.	a).	Define Controllability and Observability and explain how to calculate them using kalman's test matrix.	5	2	7
	b).	Consider the state model given below and find the transfer function of the system. $\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}2 & 3\\ -4 & -2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}3\\ 5\end{bmatrix}[u] \;;\; [y] = \begin{bmatrix}1 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$	5	3	8

[B19EC3105]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech I Semester (R19) Regular Examinations

ELECTRONIC MEASUREMENTS AND INSTRUMENTATION

DEPARTMENT OF ECE

MODEL QUESTION PAPER

TIME: 3 Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

*****

			CO	KL	M
		UNIT – I			
1.	a).	Discuss briefly the different types of static errors of a measuring instrument	CO1	K2	8
	b).	List out different AC voltmeters and explain the working of any one voltmeter in detail.	CO1	K2	7
		OR			
2.	a).	Explain the following terms in detail (i) Accuracy (ii) Resolution (iii) Precision (iv) Expected value	CO 1	K2	8
	b).	Explain the working of a true RMS voltmeter with the help of a suitable block diagram.	CO 1	K2	7
		UNIT – II			
3.	a).	What is Thermistor and explain it’s importance along with advantages of it.	CO 5	K2	7
	b).	Explain the principle of operation of strain gauges with the help of neat diagrams.	CO 5	K3	8
		OR			
4.	a).	Draw the LVDT and explain it’s operation in detail.	CO 5	K2	8
	b).	What are the modes of operation of piezo electric crystals? Explain in detail.	CO 5	K2	7
		UNIT – III			
5.	a).	Draw the Block diagram of simple CRO and explain it’s working.	CO 3	K3	8
	b).	Draw the circuit diagram of Dual trace oscilloscope and explain it’s operation in detail.	CO 3	K3	7
		OR			
6.	a).	Explain the measurement procedure of Lissajous patterns with one example.	CO 3	K3	8
	b).	Explain the concept of Digital storage oscilloscope along with circuit diagram.	CO 3	K3	7
		UNIT – IV			

7.	a).	Explain the operation of Maxwell's bridge and derive the condition for balance of a bridge.	CO 4	K3	8
	b).	Draw the circuit diagram of Schering's bridge and explain the operation of it.	CO 4	K2	7
OR					
8.	a).	Derive the equations of balance for an Anderson bridge? discuss the advantages of the bridge.	CO 4	K2	8
	b).	Draw the circuit of Wein bridge and derive the expression for bridge balance	CO 4	K3	7
UNIT – V					
9.	a).	Discuss Square wave and Pulse generator with neat block diagrams.	CO 2	K2	8
	b).	Explain the working principle of a harmonic distortion analyzer.	CO 2	K3	7
OR					
10.	a).	Illustrate the working of a function generator with a neat block diagram.	CO 2	K3	8
	b).	Draw the block diagram of random noise generator and explain with neat waveforms.	CO 2	K2	7

[B19 EC3106]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech I Semester (R19) Regular Examinations

DATA COMMUNICATIONS AND COMPUTER NETWORKS

DEPARTMENT OF ECE

MODEL QUESTION PAPER

TIME: 3 Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

*****

			CO	KL	M
		UNIT – I			
1.	a).	Explain various signal encoding methods and write NRZ codes and RZ codes for the following data streams 01011010 and 11001011.	1	2	8
	b).	Explain Synchronous and asynchronous modes of data transmission.	1	3	7
		OR			
2.	a).	Discuss various types of MODEM’s and explain with block schematic diagram DPSK modulator and demodulator	1	2	8
	b).	Compare the characteristics of different network topologies.	1	3	7
		UNIT – II			
3.	a).	Compare connection oriented and connectionless service.	2	3	7
	b).	Compare the functions of Network layer and Data link layer of OSI model in transferring data between source and destination separated by multiple networks.	2	2	8
		OR			
4.	a).	Explain statistical TDM with a neat diagram. Illustrate its advantages over synchronous TDM.	2	3	7
	b).	With the help of a timing diagram, distinguish circuit switching and packet switching techniques.	2	2	8
		UNIT – III			
5.	a).	The frame received by the receiver is 10111101100. Using the generator polynomial G(x) = + 1, verify if the frame is correct or damaged.	3	3	7
	b).	With a neat flow diagram explain the selective repeat ARQ protocol.	3	2	8
		OR			
6.	a).	Describe various CSMA protocols. Compare their throughput performance.	3	3	8
	b).	Describe the MAC sub layer frame format of IEEE standard 802.4.	3	2	7
		UNIT – IV			
7.	a).	Distinguish the characteristics of virtual circuit and Datagram subnet.	4	2	8
	b).	Differentiate the working of ethernet switch and router.	4	3	7
		OR			
8.	a).	Explain the operation of the Token Bucket Algorithm. Determine its advantages over the Leaky Bucket Algorithm.	4	3	8

	b).	Explain about IPV4 Addressing Scheme.	4	2	7
UNIT – V					
9.	a).	Explain about elements of Transport protocols.	5	2	8
	b).	Explain the operation of UDP with the help of its header format. List the applications.	5	2	7
OR					
10.	a).	Describe the main function of DNS.	5	2	8
	b).	Explain about the World wide web.	5	2	7

[B19 EC 3107]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech I Semester (R19) Regular Examinations

Digital System Design using HDL

DEPARTMENT OF ECE

MODEL QUESTION PAPER

TIME: 3 Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

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		UNIT – I	**CO**	**KL**	**M**
1.	a).	What are levels of Abstraction in VHDL?	1	2	7
	b).	Distinguish between signals and variables in VHDL.	1	2	8
		OR			
2.	a).	Explain about concurrent and sequential statements with suitable examples.	1	2	8
	b).	With the help of a block diagram, explain the program structure of VHDL.	1	2	7
		UNIT – II			
3.	a).	Realize 16:1 multiplexer with 4:1 multiplexer in VHDL programming.	1	3	8
	b).	What is FSM? Design any sequential circuit using VHDL programming.	1	3	7
		OR			
4.	a).	Design sequential 3-bit counter using VHDL programming.	1	3	8
	b).	Realize 4-bit full adder in VHDL using structural modelling.	1	3	7
		UNIT – III			
5.	a).	Explain verilog Data Types and operators.	2	2	7
	b).	Implement the verilog HDL source code and logic diagram for 1-bit full adder using dataflow style.	2	3	8
		OR			
6.	a).	How are blocking assignments different from non-blocking assignments?	2	2	7
	b).	What are the primitive gates supported by Verilog HDL? Write the Verilog HDL statements to instantiate all the primitive gates.	2	3	8
		UNIT – IV			
7.	a).	Design a negative edge-triggered D-flip flop (D_FF) with asynchronous clear, active high. Use behavioural statements only. (Hint: Output q of D_FF must be declared as reg). Design a clock with a period of 10 units and test the D_FF.	3	4	8
	b).	What are rise, fall and turn-off delays in verilog? How are they specified in verilog?	3	2	7
		OR			
8.	a).	With syntax explain conditional branching and loop statements available in Verilog HDL behavioural description.	3	2	7
	b).	write a dataflow description for 4-bit full adder	3	4	8
		UNIT – V			
9.	a).	Explain the state machine coding.	3	3	7
	b).	Describe BIST with the help of LFSR.	4	4	8
		OR			
10.	a).	Design a Modulo -8 counter using a sequential circuit approach using D-lipflop.	3	4	8
	b).	Explain scan- path technique in sequential circuit testing.	4	3	7

[B19 EC 3108]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech I Semester (R19) Regular Examinations

SOFT COMPUTING TECHNIQUES

DEPARTMENT OF ECE

MODEL QUESTION PAPER

TIME: 3 Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

*****

			CO	KL	M
		UNIT – I			
1.	a).	What is soft computing? Compare soft and hard computing?	1	2	8
	b).	explain about rules based system	1	2	7
		OR			
2.	a).	What is the significance of the AI approach? and explain briefly	1	2	8
	b).	explain intelligent control with neat architecture	1	2	7
		UNIT – II			
3.	a).	What is an activation function ?write different types of activation functions	2	3	8
	b).	Explain about Biological Neuron	2	3	7
		OR			
4.	a).	What is a neuron? Explain single layer and multilayer artificial neural networks with neat sketches?	2	3	7
	b).	What is the importance of bias in ANN? Explain different activation functions that are used in ANN?	2	3	8
		UNIT – III			
5.	a).	Write about Fuzzy relations and fuzzy rules?	3	2	8
	b).	What do you mean by a Fuzzy set? Explain about Fuzzy membership functions?	3	2	7
		OR			
6.	a).	Explain different types of Defuzzification techniques with examples?	3	3	8

	b).	Explain any two applications of Fuzzy Logic?	3	3	7
UNIT – IV					
7.	a).	Explain about Genetic Operators?	4	3	8
	b).	write about the basic framework of GA?	4	3	7
OR					
8.	a).	Explain compare and contrast Ant colony algorithms and Tabu Search algorithms?	4	3	8
	b).	Explain the concept of Genetics and Evolution?	4	3	7
UNIT- V					
9.		Explain stability analysis of neural network interconnection system	5	3	15
OR					
10.		Implement fuzzy logic controller using MATLAB fuzzy logic toolbox	5	4	15

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
III B. Tech I Semester (R19) Regular Examinations
OPERATING SYSTEMS
DEPARTMENT OF ECE

MODEL QUESTION PAPER

TIME: 3 Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT**
All questions carry equal marks

<table>
<tr><th></th><th></th><th></th><th>CO</th><th>KL</th><th>M</th></tr>
<tr><td colspan="6">UNIT-I</td></tr>
<tr><td>1</td><td>a.</td><td>Explain the abstract view of system components.</td><td>1</td><td>2</td><td>7</td></tr>
<tr><td></td><td>b.</td><td>Discuss the Simple Operating System Structure.</td><td>1</td><td>2</td><td>8</td></tr>
<tr><td colspan="6">OR</td></tr>
<tr><td>2</td><td>a.</td><td>Explain different types of Operating Systems.</td><td>1</td><td>2</td><td>7</td></tr>
<tr><td></td><td>b.</td><td>Define a System call. Explain the various types of system calls provided by Operating System,</td><td>1</td><td>2</td><td>8</td></tr>
<tr><td colspan="6">UNIT-II</td></tr>
<tr><td>3</td><td>a.</td><td>Differentiate one- to- one, many- to-one multi threading models.</td><td>2</td><td>2</td><td>7</td></tr>
<tr><td></td><td>b.</td><td>Explain Dining Philosophers problem? Discuss the solution to Dining Philosopher’s problem using monitors.</td><td>2</td><td>2</td><td>8</td></tr>
<tr><td colspan="6">OR</td></tr>
<tr><td>4</td><td>a.</td><td>Explain Primitive Priority Scheduling Algorithms with an Example?</td><td>2</td><td>2</td><td>7</td></tr>
<tr><td></td><td>b.</td><td>Discuss the solution to Reader/Writers Problem using semaphores.</td><td>2</td><td>2</td><td>8</td></tr>
<tr><td colspan="6">UNIT-III</td></tr>
<tr><td>5</td><td>a.</td><td>Differentiate paging and segmentation.</td><td>3</td><td>2</td><td>7</td></tr>
<tr><td></td><td>b.</td><td>Explain briefly the performance of Demand paging with an example.</td><td>3</td><td>2</td><td>8</td></tr>
<tr><td colspan="6">OR</td></tr>
<tr><td>6</td><td>a.</td><td>Define Page Fault. When does a page fault occur? Describe the action taken by OS when page fault occurs.</td><td>3</td><td>2</td><td>7</td></tr>
<tr><td></td><td>b.</td><td>Apply FIFO and LRU page replacement algorithms for the following string to determine the number of page faults.
7 0 1 2 0 3 0 4 2 3 0 2 1 2 0 1 7 0 1 for a memory with ‘3’ frames.</td><td>3</td><td>3</td><td>8</td></tr>
<tr><td colspan="6">UNIT-IV</td></tr>
<tr><td>7</td><td>a.</td><td>Apply the deadlock detection algorithm to determine deadlock will exist or not for the following system with 5 process and 3 resource types (resource type A has 7 instances, B has 2 instances, and C has 6 instances)
Snapshot at time T0
<table>
<tr><td>Process</td><td>Allocation</td><td></td><td></td><td>Request</td><td></td><td></td><td>Available</td><td></td><td></td></tr>
<tr><td></td><td>A</td><td>B</td><td>C</td><td>A</td><td>B</td><td>C</td><td>A</td><td>B</td><td>C</td></tr>
<tr><td>P0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>P1</td><td>2</td><td>0</td><td>0</td><td>2</td><td>0</td><td>2</td><td></td><td></td><td></td></tr>
<tr><td>P2</td><td>3</td><td>0</td><td>3</td><td>0</td><td>0</td><td>0</td><td></td><td></td><td></td></tr>
</table>
</td><td>4</td><td>3</td><td>8</td></tr>
</table>

		P3 2 1 1 1 0 0 P4 0 0 2 0 0 2			
	b.	Explain various File access methods with Suitable examples	4	2	7
OR					
8	a.	Explain deadlock avoidance using banker's algorithm with suitable example.	4	2	8
	b.	Apply FCFS, SSTF disk arm scheduling schemes to find total number head movements for the following string 98 183 37 122 14 124 65 67 assume the head pointer at 53.	4	3	7
UNIT-V					
9	a.	Explain System and Network Threats	5	2	7
	b.	Describe the System Component of Windows XP architecture	5	2	8
OR					
10	a.	Explain Principles and domain Protections.	5	2	7
	b.	Describe the components of the Linux System.	5	2	8

[B19 EC 3201]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech II Semester (R19) Regular Examinations

MICROPROCESSORS AND MICROCONTROLLERS

DEPARTMENT OF ECE

MODEL QUESTION PAPER

TIME: 3 Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

*****

			CO	KL	M
UNIT – I					
1.	With help of neat diagram explain the internal architecture of INTEL 8085 microprocessor		1	2	15
OR					
2.	a).	Draw and explain the interrupt structure of the 8085 microprocessors.	1	2	8
	b).	Draw the timing diagram for the execution of instruction **OUT 85H**	1	3	7
UNIT – II					
3.	a).	Explain the addressing modes of 8085 with examples	2	2	8
	b).	Write an 8085-assembly language program to multiply two data bytes stored in memory location 1001h and 1002h. store the result at memory location 1003h and 1004h	2	3	7
OR					
4.	a).	What is subroutine? Explain how it is implemented in 8085 using CALL and RET Instructions	2	2	8
	b).	Write an 8085-assembly language program to count the number of logical 1's and 0's in a given byte.	2	3	7
UNIT – III					
5.		With neat diagram explain the internal architecture of INTEL 8086 microprocessor	3	2	15
OR					
6.	a).	Draw the flag register of 8086 and explain the function of each flag in details	3	3	7
	b).	What is memory segmentation? Explain different segments available in 8086 microprocessors	3	2	8
UNIT – IV					
7.	a).	Draw the programmable register array of 8086 and explain the function of each Register	4	2	8
	b).	Write an 8086-assembly language program to convert 8-bit BCD number into Binary number.	4	3	7
OR					

8.	a).	Explain the different data addressing modes of 8086 Microprocessor with examples.	4	2	8
	b).	Write an 8086-assembly language program to add two four-digit BCD numbers	4	3	7
UNIT – V					
9.	a).	With the help of neat diagram, explain the Internal block diagram of 8051	5	2	15
OR					
10.	a).	Distinguish between Microprocessor and Microcontroller	5	3	7
	b).	Explain Internal RAM memory organization of 8051.	5	2	8

[B19 EC 3202]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech II Semester (R19) Regular Examinations

DIGITAL SIGNAL PROCESSING

DEPARTMENT OF ECE

MODEL QUESTION PAPER

TIME: 3 Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

*****

			CO	KL	M
		UNIT – I			
1.	a).	Find the Z-transform of the signal $x(n) = 2^n u(n) - 3^n u(-n-1)$ and its region of convergence	1	2	7
	b).	Realise the series & parallel canonical realizations of the following digital transfer function $$X(Z) = \frac{z^2 + 2z + 4}{(z-8)(z^2 - 0.9z + 0.14)}$$	1	2	8
		OR			
2.	a).	Compute the response of the system$y(n) = 0.7y(n-1) - 0.12y(n-2) + x(n-1) + x(n-2)$ to the input $\mathrm{x}(n) = u(n)$ Discuss the stability of the above DT system?	1	2	7
	b).	Find the inverse Z transform of $X(z) = \frac{z^2}{z^2 - 2rz\cos\theta + r^2}$	1	2	8
		UNIT – II			
3.	a).	Compute the DFT of the following sequence using Radix–2DIT FFT flow graph. Show the all-intermediate stage results: $x(n) = \{0,1,2,0,-2,-1,0,0\}$	2	2	7
	b).	Find the DFT of the sequence x(n) = {3, 2, 5, 4}, Using this result, find the DFT of {25, 20, 15, 10}. State the property of DFT used?	2	2	8
		OR			
4.	a).	Obtain the circular convolution of the two sequences given below using DFTd$x_1(n) = \{1,-2,3,1\}$,$x_2(n) = \{2,3,0,-4\}$	2	2	8
	b).	Compare in place computation and natural input-natural output computation methods. Discuss the computational complexities involved in direct DFT and FFT.	2	2	7
		UNIT – III			
5.	a).	Compare Chebyshev and Butterworth analogfilters?	3	2	7
	b).	Design digital Butterworth lowpass IIR filter using BLT method. The filter specifications are given by i) -3dB cut-off frequency at 0.5π rad, ii) at least 15dB attenuation at 0.75π rad	3	3	8
		OR			

6.	a).	Compare Impulse invariance and Bilinear transformation methods of IIR digital filter design.	3	2	8
	b).	Convert the following analog filter with transfer function using impulse invariance method $$H_a(S) = \frac{s + 0.2}{(s + 0.2)^2 + 25)}$$	3	3	7
		UNIT – IV			
7.	a).	Design a linear-phase low pass FIR digital filter to meet the following specifications: (i) Pass band = 0 to 10 kHz (ii) Sampling frequency = 100 kHz(iii) Filter order =10. Compute the impulse response of the desired FIR digital filter using Hamming window	4	3	9
	b).	What is Gibb's phenomenon? Discuss the selection criteria of windows with respect to FIR filter design	4	2	6
		OR			
8.	a).	Show that FIR filters provide constant group delay and phase delay?	4	2	7
	b).	Design a linear-phase band pass FIR digital filter to meet the following specifications: (i) Pass band = 100Hz to 200Hz (ii) Sampling frequency = 1000Hz (iii) No. of samples =11. Compute the impulse response of the desired FIR digital filter using Rectangular and Hamming windows.	4	3	8
		UNIT – V			
9.	a).	Explain how Sub band coding of speech signals reduces the bit rate	5	2	7
	b).	Illustrate the operation of up-sampler, down-sampler, Interpolator and Decimator in time and frequency domains with neat sketches	5	2	8
		OR			
10.	a).	What is a digital filter bank? Describe its operation & applications	5	2	7
	b).	Write short notes on the finite precision arithmetic effects in the realization of digital filters.	5	2	8

[B19 EC 3203]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech II Semester (R19) Regular Examinations

VLSI DESIGN

DEPARTMENT OF ECE

MODEL QUESTION PAPER

TIME: 3 Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

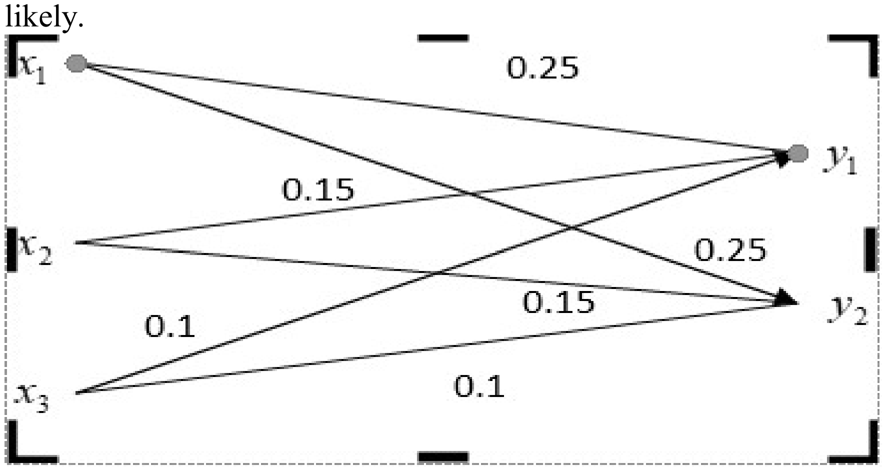
*****

		UNIT – I	**CO**	**KL**	**M**
1.	a).	Explain the NMOS fabrication steps with neat diagrams.	1	3	8
	b).	Discuss alternate forms of Pull-up Configurations and derive the relation between pull –up to pull-down ratio for nMOS inverter.	1	3	7
		OR			
2.	a).	With neat diagrams explain the process of P-well CMOS Inverter.	1	3	7
	b).	Compare CMOS ,BiCMOS and Bipolar technologies.	1	3	8
		UNIT – II			
3.	a).	Draw the stick diagrams and layouts for (a) nMOS inverter (b) CMOS inverter (c) 3 Input NAND and NOR gates	2	3	8
	b).	Define Buried contact, Butting contact and Via contact.	2	3	7
		OR			
4.	a).	Sketch λ-based design rules for wires, transistors and contacts.	2	3	8
	b).	Draw the layout diagram for OAI logic using CMOS.	2	3	7
		UNIT – III			
5.	a).	What is meant by Delay unit? Estimate NMOS inverter pair delays.	3	2	8
	b).	Explain limitations of scaling.	3	2	7
		OR			
6.	a).	Draw scaled NMOS transistor and derive all scaling factors for device parameters. Consider Combined V and D scaling model	3	2	8
	b).	Calculate total on resistance of CMOS inverter where $Z_{PU}/Z_{PD}=4/1$	3	2	7
		UNIT – IV			
7.	a).	Explain Master-Slave (edge-triggered) S-R flip-flop	3	3	8
	b).	Distinguish between combinational and sequential switching circuits	3	2	7
		OR			
8.	a).	Explain charge leakage and charge sharing in dynamic logics.	3	2	8
	b).	Give a brief explanation about Latches and registers.	3	2	7
		UNIT – V			
9.	a).	With the help of logic diagram, explain about Output response analyser using LFSR.	4	3	8
	b).	Write various steps to be followed for test mode in Scan Design Techniques?	4	3	7
		OR			
10.	a).	Explain about various Scan design techniques.	4	3	8
	b).	Explain about controllability and observability?	4	3	7

[B19 EC 3204]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
III B. Tech II Semester (R19) Regular Examinations
INFORMATION THEORY AND CODING (PE-2)
MODEL QUESTION PAPER

TIME: 3 Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT**
All questions carry equal marks
*****

			CO	KL	M
		UNIT-I			
1.	**a).**	Discuss in detail about Shannon's source coding theorem.	**CO1**	**K2**	**7**
	b).	An analog signal is band limited to 800 Hz, sampled at the Nyquist rate, and the samples are quantized into four levels. The quantization levels are assumed independent and occur with probabilities (1/8, 1/8, 3/8, and 3/8). Find the entropy H(X) and information rate R of the source	**CO1**	**K3**	**8**
		OR			
2.	**a).**	Discuss the steps involved in the Shannon-Fano algorithm.	**CO2**	**K2**	**7**
	b).	A source emits messages with probabilities (1/2, 1/4, 1/8, 1/16, 1/32, and 1/32). Calculate (i) entropy of the source, (ii) Apply Shannon-Fano algorithm to devise a binary code for this source and find its coding efficiency and redundancy.	**CO2**	**K3**	**8**
		UNIT-II			
3.	**a).**	Find the mutual entropy H(X;Y) for the channel shown below. Joint probabilities P(XY) are given. Assume source symbols are equally likely. 	**CO3**	**K4**	**8**
	b).	Derive the expressions for mutual entropy, mutual information and joint probabilities	**CO3**	**K3**	**7**
		OR			

4.	**a).**	State & explain Shannon's noisy channel coding theorem. Find the channel capacity of a binary symmetric channel.	**CO3**	**K3**	**7**
	b).	Derive the expression for the channel capacity of AWGN channel.	**CO3**	**K3**	**8**
		UNIT-III			
5.	**a).**	Explain the generation of a linear systematic (n, k) block code using generator matrix. Define minimum Hamming distance d_{min} of a code. What is the relation between d_{min} and the error correcting capacity of a code	**CO4**	**K2**	**7**
	b).	Find the generator polynomial & parity check polynomial for a linear (7,4) systematic cyclic code. Use them to code and decode a message 1010.	**CO4**	**K3**	**8**
		OR			
6.	**a).**	Demonstrate that (7,4) Hamming code can correct a single error & detect a double error by syndrome decoding	**CO4**	**K3**	**8**
	b).	Write about BCH codes & CRC codes.	**CO4**	**K2**	**7**
		UNIT-IV			
7.	**a).**	Draw the structure of a rate 1/2 Convolutional coder for g_1=[1 0 1] and g_2=[011]. Explain the encoding process. Construct the state diagram, trellis diagram & code tree. Find the coder output for input data = [1 0 10 1].	**CO5**	**K3**	**8**
	b).	Write in detail about structural properties of convolution codes.	**CO5**	**K2**	**7**
		OR			
8.	**a).**	Distinguish between exhaustive search method and Viterbi decoding of Convolutional codes. Explain how the decoding complexity increases with the constraint length?	**CO5**	**K4**	**8**
	b).	Write in detail about coding for compound error channels	**CO5**	**K2**	**7**
		UNIT-V			
9.	**a).**	What is a MIMO system? Explain Space-Time Coded MIMO system using 2-transmit Alamouti STBC code.	**CO5**	**K2**	**8**
	b).	Write in detail about concatenated codes	**CO5**	**K2**	**7**
		OR			
10.	**a).**	Describe Turbo codes structure, encoding & advantages.	**CO5**	**K2**	**8**
	b).	Write short notes on: LDPC codes, Tanner graphs & applications.	**CO5**	**K2**	**7**

[B19 EC 3205]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech II Semester (R19) Regular Examinations

Digital IC Design -----PE2

DEPARTMENT OF ECE

MODEL QUESTION PAPER

TIME: 3 Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

*****

			CO	KL	M
		UNIT – I			
1.	a).	Explain the operation of a CMOS inverter and derive the expressions for fall and rise times	1	K3	8
	b).	Design a pseudo-NMOS logic gate that realizes the function $\square\square\square = x1(\square 2 + \square 3)$. Give reasonable dimensions assuming Lmin = 0.5μm	2	K3	7
		OR			
2.	a).	Discuss about the following parameters of pseudo nMOS Logic a)Inverter threshold voltage b)Output high voltage	1	K3	8
	b).	Draw and explain the operation of a 2-input NMOS NAND gate circuit.	2	K3	7
		UNIT – II			
3.	a).	Explain the operation of a CMOS full adder circuit with a neat diagram.	3	K3	8
	b).	Realizing Boolean expressions$\square\square\square = \square 1.\square 2(\square 4 + \square 3)$ using NMOS gates	3	K4	7
		OR			
4.	a).	Design a NAND gate using Transmission gates.	3	K4	8
	b).	Explain the operation of pass transistor logic and explain leakage and subthreshold currents in dynamic pass gate.	3	K3	7
		UNIT – III			
5.	a).	Compare static and dynamic latches.	3	K3	8
	b).	Draw the logic diagram and truth table of a CMOS D-Latch and explain its operation.	3	K3	7
		OR3			
6.	a).	Draw the logic diagram and truth table of a CMOS clocked SR flip-flop and explain its operation.	3	K3	8
	b).	Explain the behaviour of bistable elements	3	K3	7
		UNIT – IV			
7.	a).	List out the different types of issues in CMOS dynamic logic design? Explain anyone with a neat sketch.	3	K3	8
	b).	Explain the operation dynamic CMOS transmission gate	3	K3	7
		OR			
8.	a).	Write short notes on voltage Bootstrapping	3	K1	8
	b).	Write short notes on the performance of Dynamic CMOS circuits	3	K1	7
		UNIT – V			

9.	a).	Discuss about various leakage currents in SRAM	4	K1	8
	b).	Write short notes on RAM Array Organization	4	K1	7
OR					
10.	a).	Discuss about variousLeakage currents in DRAM cell	4	K1	8
	b).	With a neat sketch? Explain the principle of NOR gate flash memory	4	K3	7

[B19 EC 3206]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech II Semester (R19) Regular Examinations

DIGITAL SIGNAL PROCESSORS AND ARCHITECTURE (PE-2)

DEPARTMENT OF ECE

MODEL QUESTION PAPER

TIME: 3 Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

*****

			CO	KL	M
		UNIT – I			
1.	a).	Explain the basic Digital signal processing system with the help of block diagram.	1	2	7
	b).	Explain in detail about sources of errors in DSP implementations.	1	2	8
		OR			
2.	a).	Discuss about the DFT and FFT calculations in Digital signal processors.	1	2	8
	b).	Write about fixed point and floating point formats.	1	2	7
		UNIT – II			
3.	a).	Discuss in brief about the data addressing capabilities of programmable DSP devices with examples.	2	2	7
	b).	Explain the operation of address generation unit in a DSP processor with neat block diagram.	2	3	8
		OR			
4.	a).	What are the features for external interfacing in a DSP processor.	2	3	7
	b).	Write about the programmability and program execution sequence of a DSP	2	3	8
		UNIT – III			
5.	a).	What is the need an interrupt in a processor? Write about Interrupts of TMS320C54XX Processor.	3	3	8
	b).	Explain the concept of Pipelining for speeding up the execution of an Instruction.	3	3	7
		OR			
6.	a).	Write about different on chip peripherals of TMS320C54XX Processor.	3	2	7
	b).	Explain about the memory space organization of TMS320C54XX Processor.	3	2	8
		UNIT – IV			
7.	a).	Draw and explain the block diagram of memory interface for TMS320C5416 processor.	4	3	7
	b).	How does DMA help in increasing the processing speed of a DSP processor?	4	4	8
		OR			
8.	a).	How an external bus can be get interfaced with TMS320C54XX processor? Explain.	4	4	8
	b).	Explain the Parallel I/O and Programmable I/O.	4	2	7

UNIT – V					
9.	a).	Explain in detail about Decoding Scheme of PPM Receiver.	**5**	**2**	**8**
	b).	How the speech being processed through DSP Processor? Explain.	**5**	**2**	**7**
OR					
10.	a).	Explain about JPEG encoding.	**5**	**2**	**8**
	b).	Write about the implementation of Biometric Receiver.	**5**	**2**	**7**

[B19 EC 3207]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

III B. Tech II Semester (R19) Regular Examinations

DATABASE MANAGEMENT SYSTEMS(PE-2)

DEPARTMENT OF ECE

MODEL QUESTION PAPER

TIME: 3 Hrs. **Max. Marks: 75 M**

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

*****

		UNIT-I	**CO**	**KL**	**M**
1	a.	Compare Database Management Systems with File Processing Systems	1	2	7
	b.	Explain the roles of different database users	1	2	8
		OR			
2	a.	Discuss the applications of Database Management Systems	1	2	7
	b.	Describe the structure of a Database Management System	1	2	8
		UNIT-II			
3		Give syntax and apply the SQL commands for defining two example tables of your choice. Then insert data, update data in the tables	2	3	15
		OR			
4		What are relational instances and schemas? How'd you use keys and schemas in relational model?	2	3	15
		UNIT-III			
5	a.	Apply conceptual DB design and draw E-R diagram for the following situations by assuming appropriate Attributes i) A Part is supplied by many suppliers at different costs and a supplier supplies many partsii) An employee works in at most one department and a departmenthasmany employeesiii) A house has at least and at most one owner and ownerhasmany houses iv) A Muslim woman marries at most one man and a Muslim man could marry many woman	3	3	8
	b.	Demonstrate set operations in SQL	2	3	7
		OR			
6	a.	Apply different kinds of joins in SQL to example queries	2	3	8
	b.	Illustrate basic features of ER model	3	3	7
		UNIT-IV			
7	a.	Apply Loss-less join decomposition into BCNF for an example table	4	3	8
	b.	Apply dependency preserving decomposition into 3NFfor an example table	4	3	7
		OR			
8		Illustrate Normal forms from 1 NF to BCNF.	4	3	15
		UNIT-V			
9		What are page tables and Transaction tables? Describe analysis, redo and undo steps of ARIES.	5	2	15
		OR			
10		Explain 2PL and time stamp ordering protocols	5	2	15